

Questionnaire for Evaluating AI-Generated Images

Study 3: Evaluation of AI-Generated Images

Participants completed the following questionnaire after generating design images using AI-assisted tools. The questionnaire consisted of four sections: (1) participant background, (2) prior experience with traditional rendering and AI tools, (3) evaluation of AI-generated images, and (4) post-task reflections. Unless otherwise specified, all evaluation items employed a five-point Likert scale (1 = Very Poor, 5 = Excellent).

Section A. Participant Background

A1. Participant Name: _____

A2. Email Address: _____

A3. Course Enrollment

☐ URPN 420

☐ ARCH 205

A4. Years of Design Training

(Number of years since your first architecture, landscape architecture, or urban design project.)

Section B. Previous Experience

B1. Before this course, have you used traditional rendering methods (e.g., SketchUp, Rhino, Revit, V-Ray, Lumion, Enscape, Photoshop, or Illustrator) to create design images?

☐ Yes

☐ No

B2. If yes, how would you rate your experience with traditional rendering methods?

1	2	3	4	5
Very Dissatisfied	Dissatisfied	Neutral	Satisfied	Very Satisfied

B3. Have you previously used AI-generative tools for image creation?

- ☐ No
- ☐ ChatGPT, Copilot, or other text-based AI applications
- ☐ Midjourney
- ☐ Stable Diffusion
- ☐ SketchUp Diffusion
- ☐ Other: _____
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B4. Which aspects do you consider the primary advantages of AI-generated images? (Select all that apply.)

- ☐ Cost
- ☐ Tool usability
- ☐ Speed of image generation
- ☐ Quality and clarity of generated images
- ☐ Image customizability
- ☐ Other: _____
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Section C. Evaluation of AI-Generated Images

Please evaluate the AI-generated images based on the following criteria.

Scale

Rating	Description
1	Very Poor
2	Poor
3	Neutral
4	Good
5	Excellent

C1. Image Realism

Evaluation Item	1	2	3	4	5
Overall image realism	☐	☐	☐	☐	☐
Material representation	☐	☐	☐	☐	☐
Scale accuracy	☐	☐	☐	☐	☐
Lighting and shadow	☐	☐	☐	☐	☐

C2. Design Rationality

Evaluation Item	1	2	3	4	5
Integration with surrounding context (e.g., landscape elements and nearby structures)	☐	☐	☐	☐	☐
Layout and functionality	☐	☐	☐	☐	☐
Meeting prompt requirements	☐	☐	☐	☐	☐

C3. Adherence to Design Instructions

Evaluation Item	1	2	3	4	5
Reflection of the intended design style	☐	☐	☐	☐	☐
Logical and practical coherence	☐	☐	☐	☐	☐
Accuracy of key design elements	☐	☐	☐	☐	☐
Overall aesthetic appeal	☐	☐	☐	☐	☐

Section D. Future Adoption and Reflection

D1. Would you consider using AI-generated image tools in your future design projects?

☐ Yes

☐ No

D2. Please share your reflections on using AI-generated images for this project.

Data availability

The de-identified responses to this questionnaire, together with the AI-generated images they refer to, are available at <https://github.com/suririri/genai-design-renderings>.